



MODULE CONTRACT: Core Types

Module ID: core-types

Module Name: Core Types

Module Type: package

Version: 1.0.0

Date: 2026-06-13

Status: active

Authority Type: code

1. ☒ MODULE IDENTITY

Field	Value
moduleId	core-types
moduleName	Core Types
moduleType	package
moduleStatus	active
authorityType	code
package	@balloo/core-types
version	1.0.0

2. ☒ PURPOSE

Problem Solved:

Centralized type definitions — Without core-types, each package and app defines its own types, leading to duplication and inconsistency.

Why Module Exists:

- ☒ Single source of truth for TypeScript types
- ☒ Shared type definitions across monorepo
- ☒ Type safety and consistency
- ☒ Reduced maintenance burden

Scope:

- ☒ TypeScript type definitions
- ☒ Interface definitions
- ☒ Type utilities
- ☒ Generic types

Out of Scope:

- ☐ Runtime code
- ☐ Business logic
- ☐ API implementations
- ☐ UI components

3. ☒ ARTIFACTS

Code Artifacts:

```
packages/core-types/  
├── src/  
│   ├── index.ts           # Main entry point  
│   └── types/  
│       ├── common.ts      # Common types (ID, Timestamp, Status)  
│       ├── node.ts        # Node-related types  
│       ├── messenger.ts   # Messenger types  
│       ├── admin.ts       # Admin types  
│       └── index.ts        # Type exports  
├── interfaces/  
│   ├── NodeConfig.ts      # Node configuration interface  
│   ├── Message.ts         # Message interface  
│   └── index.ts           # Interface exports  
├── package.json  
├── tsconfig.json  
└── README.md
```

Doc Artifacts:

- ☒ README.md — Package documentation
- ☒ MODULE_SUMMARY_core-types.md — Human-readable summary
- ☒ MODULE_CONTRACT_core-types.md — This contract
- ☒ API documentation (generated)

Contract Artifacts:

- ☒ This module contract
- ☒ Type specifications
- ☒ Interface definitions

Data Artifacts:

- ☐ None — types only, no runtime data

Runtime Artifacts:

- ☐ None — no runtime code

4. INTERFACES

Public Interfaces:

Type Exports:

```
// Common
export type ID = string;
export type Timestamp = number;
export type Status = 'active' | 'inactive' | 'pending';

// Node
export interface NodeConfig {
  nodeId: ID;
  nodeName: string;
  nodeType: 'laptop' | 'server' | 'nas' | 'aio';
  // ...
}

export interface NodeState {
  nodeId: ID;
  status: Status;
  lastSeen: Timestamp;
  // ...
}

// Messenger
export interface Message {
  messageId: ID;
  conversationId: ID;
  senderId: ID;
  content: string;
  timestamp: Timestamp;
  // ...
}

export interface Conversation {
  conversationId: ID;
  participants: ID[];
  lastMessage: Message;
  // ...
}
```

Internal Interfaces:

- ☒ None — all interfaces are public

Endpoint Surface:

Import Surface:

```
import { MessageType, NodeConfig } from '@balloo/core-types';
```

No HTTP/RPC endpoints — package module.

Docs Entrypoints:

- ☒ MODULE_SUMMARY_core-types.md
- ☒ MODULE_CONTRACT_core-types.md
- ☒ packages/core-types/README.md

5. ☒ PLACEMENT

Repo Placement:

```
packages/core-types/
```

Node Presence:

Node	Presence	Description
laptop_control	import	Used in development
work_server	import	Used in build/deployment

Domain Exposure:

None — package module.

Distribution: npm package (@balloo/core-types)

Branch Relevance:

- ☒ main — stable version
- ☒ feature/* — development branches
- ☒ All branches that modify types

6. ☒ DEPENDENCIES

Depends On Modules:

- ☒ None — base types package

External Dependencies:

```
{
  "devDependencies": {
    "typescript": "^5.0.0"
  }
}
```

Upstream Contracts:

- ☒ None

Downstream Consumers:

Module	Type	Usage
core-config	package	Type definitions
core-i18n	package	Type definitions
core-theme	package	Type definitions
core-brand	package	Type definitions
core-ui	package	Type definitions
core-docs-schema	package	Type definitions
messenger	hybrid	Type definitions
admin-portal	hybrid	Type definitions
desktop	hybrid	Type definitions
mobile	hybrid	Type definitions
android-service	service	Type definitions

7. ☒ GENERATION RELEVANCE

Used For Codegen:

YES — HIGH Relevance

Codegen Usage:

- ☒ Type generation templates
- ☒ Interface scaffolding
- ☒ Type-safe code generation
- ☒ Contract-based type definitions

Used For Docgen:

YES — HIGH Relevance

Docgen Usage:

- ☒ API documentation generation
- ☒ Type reference docs
- ☒ Interface documentation

Reconstruction Value:

HIGH — Essential for monorepo consistency.

Audit Value:

HIGH — Types affect all dependent modules.

8. ☒ EVIDENCE

Inferred From:

- ☐ Not inferred — implemented module

Canonical Docs:

- ☒ MODULE_SUMMARY_core-types.md
- ☒ MODULE_CONTRACT_core-types.md
- ☒ packages/core-types/README.md

Related Packages:

- ☒ @balloo/core-types (npm)

Related Apps:

- ☒ messenger
- ☒ admin-portal
- ☒ desktop
- ☒ mobile
- ☒ android-service

Related Contracts:

- ☒ ModuleTypesContract.md
 - ☒ ModuleDependencyContract.md
-

9. ☒ INVARIANTS

Required Invariants:

1. ☒ **Type consistency** — All types must be consistent across monorepo
2. ☒ **No breaking changes** — Backward compatibility required
3. ☒ **Single source** — Types defined only in core-types
4. ☒ **Type safety** — All exports must be properly typed

Forbidden Assumptions:

- ☒ Don't assume types are optional — they are required
- ☒ Don't duplicate types in other packages
- ☒ Don't modify types without updating all consumers

Compatibility Notes:

- ☒ Semver versioning
- ☒ Breaking changes require major version bump
- ☒ Deprecation path for old types

10. ☒ STATUS NOTES

Implementation Status:

☒ **Implemented (v1.0.0)**

Evidence:

- ☒ Package exists in packages/core-types/
- ☒ package.json present
- ☒ Source files present (src/index.ts)
- ☒ Imported by multiple modules
- ☒ [CODEGEN] 2026-06-14 — Full type generation complete

Codegen History:

Date	Version	Changes
2026-06-14	1.0.0	Full type generation from MODULE_CONTRACT
2026-06-14	1.0.0	Added Node types (NodeConfig, NodeState, NodeTree)
2026-06-14	1.0.0	Added Admin types (AdminUser, AuditLogEntry, SystemMetrics)
2026-06-14	1.0.0	Added Type Utilities (10 utility types)
2026-06-14	1.0.0	Enhanced JSDoc documentation

Migration Notes:

- ☐ No migration needed — stable module
- ☒ Backward compatible with previous versions

Future Work:

- ☐ Add more domain-specific types (as needed)
- ☒ Type documentation enhanced with JSDoc
- ☐ Add type tests (pending test framework setup)

 METRICS

Metric	Value
Type Count	50+
Interface Count	30+
Dependent Modules	11
Import Count	100+
Codegen Relevance	HIGH
Docgen Relevance	HIGH

☒ ACCEPTANCE CRITERIA

Контракт считается выполненным если:

1. ☒ Module identity defined
2. ☒ Purpose documented
3. ☒ Artifacts listed
4. ☒ Interfaces specified
5. ☒ Placement documented
6. ☒ Dependencies listed
7. ☒ Generation relevance defined
8. ☒ Evidence provided
9. ☒ Invariants specified
10. ☒ Status notes included

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Версия: 1.0.0

Статус: Active

Автор: Koda (NLP-Core-Team)

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